

qicert: Certified Quantum-Inspired Compression of Vision–Language–Action Models with Exact Bounds, Provable Pruning, and Formal Safety Tails

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Code: <https://github.com/ZakLr/qicert> Reproduces every table via `qicert.bench.all`

Abstract—Vision–Language–Action Models (VLAMs) at 7B–10B parameters are too large for embedded vehicle and robot controllers and too expensive to certify. We present *qicert*, a backbone-agnostic pipeline that compresses VLAMs into quantized tensor-train (QTT) networks before the heaviest training, compiles cross-modal interaction blocks into exactly-evaluable commuting-Pauli families, and certifies what remains with a three-layer certificate stack: exact algebraic bounds (Lipschitz products of QTT cores; per-family pruning errors), local formal proofs (SOS boxes; Lyapunov-margin survival curves), and statistical tail bounds on signal-temporal-logic robustness estimated by iterative quantum amplitude estimation raced against the strongest honest classical estimator (RESTART+GEV) and closed by scenario-optimization and conformal calibration. A syndrome-shadow monitor guards the action stream with theorem-budgeted false-alarm rates. The INT8 baseline can match our accuracy at matched compression; it cannot compute our certificates, which are exact functions of the compressed cores.

Index Terms—tensor networks, quantized tensor train, Lipschitz certificates, Pauli grouping, Clifford diagonalization, quantum amplitude estimation, signal temporal logic, classical shadows, VLAM, autonomous driving, robotics, formal safety

I. INTRODUCTION

VLAMs fuse camera, LiDAR, radar, and language inputs into planning and control outputs. Reference models such as Alpamayo-R1 [1] and π_0 [2] demonstrate the category’s potential, but their scale makes them prohibitively expensive to train and nearly impossible to deploy on embedded hardware within the 100 ms real-time budget while satisfying functional-safety certification (ISO 26262; IEC 62061 / ISO 10218).

The challenge asks for a reproducible quantum-enhanced or quantum-inspired component with a clear advantage over a classical baseline across four bottlenecks: model footprint, attention complexity, RL alignment cost, and control-safety guarantees. Quantum-inspired tensor networks have demonstrated strong LLM compression [3], and provably efficient quantum algorithms promise gains in the high-dimensional linear algebra central to VLAM training [4]. Our answer is a *system*: one pipeline in which five quantum-inspired primitives — tensor networks, Pauli observables, amplitude encoding, syndrome measurement, and shadow tomography — are the type system of every stage (train $\rightarrow$ compress $\rightarrow$ compile $\rightarrow$ certify $\rightarrow$ monitor). The contributions are:

- 1) **Exact-certificate compression**: QTT cores double as the objects of an exact global Lipschitz bound — exact per-layer factors, residual gap measured (Sec. III) — giving a certified-safe-set column that no quantization-based baseline can produce at any compression ratio.
- 2) **Compilation-as-identity**: cross-modal interaction tensors are grouped into commuting Pauli families, simultaneously diagonalized by single Cliffords, and evaluated exactly; dropping a family costs an exact, printed error $\|\sum_{a \in F_j} c_a\|_1$ (Sec. IV).
- 3) **Quadratic query-complexity savings on the safety tail (realized oracle)**: the certified failure probability is estimated by iterative quantum amplitude estimation (IQAE) with Bayesian credible intervals, over a structured scenario domain with a closed-form certificate-margin oracle (Sec. V), raced end-to-end against RESTART+GEV at matched total cost and closed by Campi–Garatti scenario-optimization and conformal prediction.
- 4) **A runnable, self-reproducing artifact**: *qicert* (“certified quantum-inspired compression”) reproduces every table from a clean environment and emits its own hardware pathway (qubits, depth, shots).

All quantum components are simulated per challenge constraints (Sec. 5.3); resource assumptions are stated per algorithm. Results below are reported as pre-registered targets with measured values pending (Sec. IX).

II. SYSTEM ARCHITECTURE

The pipeline is identical across both tracks; only the slice specification differs.

Training time (offline, Kaggle 2×T4): Stage 0 baseline fine-tune (QLoRA/LoRA) of the backbone (AD: LLaVA-1.5-7B with Alpamayo-adjacent action head, justified per Sec. 5.4; robotics: MiniVLA-class head). **Stage 1** compress into {TT, QTT} via TT-cross (query-based, no global SVD); the safety-budgeted bond allocator sets per-layer ranks. **Stage 2** ALS/DMRG-native fine-tuning on the compressed manifold — the dense matrix is never formed. **Stage 3** compile cross-modal interaction blocks into commuting-Pauli families. **Stage 4** certify (three layers, Sec. III). **Stage 5** package into *qicert*; the bench suite reproduces every table.

TABLE I
THE SEVEN FUSION JOINTS. EACH COMPONENT CONSUMES ANOTHER’S OUTPUT, SO ABLATIONS ISOLATE CLEANLY AND THE MARGINAL COST OF EVERY EXTRA CLAIM IS NEAR ZERO.

Joint	What it buys
1	QTT cores = certificate inputs (zero extra compute)
2	Compiled families = monitor’s measurement basis (zero extra design)
3	Certificate margins = the STL predicates IQAE estimates (compositional)
4	Sec. 5.2 seeded scenarios = the amplitude-estimation oracle
5	Worst-case long-tail margin = bond-allocation objective (tail design)
6	Compiled qubit/depth/shots table = the hardware pathway
7	qicert carries everything

Stage 0: baseline fine-tune (QLoRA/LoRA)
↓ Stage 1: {TT,QTT} TT-cross compression + bond allocator
↓ Stage 2: ALS/DMRG-native fine-tuning (dense matrix never formed)
↓ Stage 3: compile commuting-Pauli families (exact, prunable)
↓ Stage 4: certify — L1 exact Lipschitz / L2 SOS + Lyapunov / L3 IQAE tails
↓ Stage 5: package qicert; bench suite reproduces every table

Fig. 1. Training-time pipeline (Sec. II); the runtime view adds the syndrome-shadow monitor (Sec. VI).

Runtime (deployed, ≤ 100 ms profile): sensor + instruction encoders $\rightarrow$ compiled interaction kernels (diagonal, $O(N \cdot k)$ -style) $\rightarrow$ QTT-compressed action head with certified margins $\rightarrow$ syndrome-shadow monitor (PDU hygiene + shadow statistics) $\rightarrow$ safe output or certified fallback action.

Fusion joints (why this is a system, not a portfolio):

Unifying principle. A VLAM that will be certified and deployed on embedded hardware is, at every stage, a computation on *structured tensors* — weights, interaction tensors, belief and robustness distributions. Quantum information is the discipline that made structured tensors a first-class programming model: tensor networks, Pauli observables, amplitude encoding, syndrome measurement, shadow tomography. We build the entire train $\rightarrow$ compress $\rightarrow$ compile $\rightarrow$ certify $\rightarrow$ monitor pipeline from those five primitives. Quantum-inspiration is the type system of the whole submission, not a label on one component.

III. THE CERTIFICATE STACK

Three layers answer three questions and are never confused.

A. Layer 1: Exact algebraic bounds

For a TT/QTT layer with cores $G_1 \dots G_d$ the reconstructed weight matrix is $W = \text{contract}(G_1, \dots, G_d)$, and norm submultiplicativity gives

$$\|W\|_2 \leq \prod_{i=1}^d \|G_i\|_2, \quad (1)$$

each core norm computed by power iteration on the small core in $O(\sum_i r_i^2 n_i)$. The bound is deliberately conservative; the *exact* per-layer operator norm is available at the same cost class: power iteration on the TT contraction performs

[Measured in N3/N5: certified-safe-set % vs compression ratio]

qicert-QTT: monotone decay, bond-spectrum predictor band
INT8 / SVD: not computable (flat 0% line, stated honestly)

Fig. 2. The asymmetry plot (pre-registered): the certificate column that quantization baselines structurally cannot produce.

matvecs $x \mapsto Wx$ in $O(\sum_i r_{i-1} n_i^2 r_i)$ per step (standard TT contraction order; $O(dn^2 r^2)$ at uniform mode size) without ever forming W . Chaining through the network with 1-Lipschitz activations (ReLU et al.),

$$L(\tilde{F}) = \prod_{\text{layers } l} \|W^{(l)}\|_2, \quad (2)$$

where each factor is the exact per-layer norm (power iteration), not the core-product bound. The residual gap between $L(\tilde{F})$ and $\sup_x \|\nabla \tilde{F}(x)\|_2$ — activation interleaving and non-orthogonality across layers — is *measured*, not assumed: N3 registers the tightness ratio $\kappa = L(\tilde{F})/\hat{L}_\nabla$, where $\hat{L}_\nabla$ estimates $\sup_{x \in B} \|\nabla \tilde{F}(x)\|_2$ by power iteration on the full TT network at adversarial sample points in B , and prints it at every Pareto point. With the input box B from the slice spec this yields the *exact safe-set certificate*

$$\text{Safe}(B) = \{x \in B : \|\tilde{F}(x) - \tilde{F}(x_{\text{ref}})\| \leq L(\tilde{F}) \text{diam}(B) < \text{margin}\}. \quad (3)$$

Non-emptiness is a *measured* column (Table V), never an assumption: a conservative L shrinks the column but cannot falsify it, and the column still separates us from INT8, for which no certificate object exists at any ratio. Its usefulness is gated by R2; where the global bound is loose, Layer 2a supplies the tight local boxes.

Theorem 1 (Lipschitz product, layer 1). *For a TT layer with cores $G_1 \dots G_d$, Eq. (1) holds with no sampling and no input-distribution assumption, and the tighter exact factor $\|W\|_2$ is computable by power iteration on the TT contraction at cost $O(k_{\text{it}} \sum_i r_{i-1} n_i^2 r_i)$ with k_{it} iterations, the dense matrix never formed. Both properties are preserved for QTT (log-mode) reshapes because they depend only on the contraction, not on mode sizes.*

The load-bearing comparison: INT8 quantization error is data-dependent (clip ranges, per-channel scales) and unfactorizable — no exact global constant exists. SVD truncation gives no certificate object to compose. Our compressed cores *are* the certificate.

B. Layer 2: Local formal proofs

(a) SOS-certified boxes. Composition polynomials of TT-structured networks admit moment-SOS hierarchies with tractable SDPs [8]. SOS is deliberately *local by construction*: each certificate covers one sampled long-tail box B_k and a low-degree (≤ 4) composition polynomial over a small slice (state dimension $\lesssim 16$), never a whole-network SDP — formal verification of a 7B network in one SDP is out of scope for

TABLE II
COMPILER PASSES. PASS 4 IS AN IDENTITY, NOT AN APPROXIMATION.

Pass	Operation	[Measured in N7: predicted $\left\ \sum_{a \in F_j} c_a \right\ _1$ vs measured Δ]	
		Target: correlation ≥ 0.95 (E-comp-2): per-family leave-one-out	Output
1	Extract interaction table $\{(c_a, P_a)\}$	Fig. 3. Predicted vs measured pruning error — the additional column no other submission can provide	Pauli table
2	Group into mutually commuting families $\{F_1, \dots, F_m\}$ (greedy coloring, polynomial; m measured at N7)		family partition
3	Per family: single Clifford C_j with $C_j^\dagger P C_j \in \{I, Z\}^{\otimes k}$		Clifford table
4	Rewrite exactly $T = \sum_j C_j^\dagger D_j C_j$, D_j diagonal		executable kernels
5	Emit hardware table: qubits = $\lceil \log_2(\text{register}) \rceil$, depth = m , shots $\leq \gamma(\ c\ _1, \epsilon)$		safe set survives a Sec. 4.2 pathway
6	Emit pruning certificates: dropping F_j costs exactly $\left\ \sum_{a \in F_j} c_a \right\ _1$		ablation knob

any method, and the hierarchy’s state-lifting structure keeps each box small. R3 budgets each box at <2 h and demotes SOS to a report line if violated. For each sampled long-tail box B_k we certify

$$\forall x \in B_k : \text{margin}(x) \geq m_k \geq 0, \quad (4)$$

proven by an exported SOS certificate (machine-checkable Gram matrix). Layer 1 is global-but-loose; Layer 2a is local-but-tight; disagreement between them is a surfaced bug, not a hidden number.

(b) Lyapunov-margin degradation curve + predictor.

The old binary “certificate survives” claim is upgraded to a continuous object. At each compression ratio r ,

$$\text{survival}(r) = \frac{\#\{\text{trajectories where } V\text{-decrease holds}\}}{\#\{\text{trajectories}\}}, \quad (5)$$

with a regression from (bond spectra, entropy signals, layer sensitivities) to $\text{survival}(r)$. The submission prints the *predicted* largest safe ratio r^* *before* measuring it, then reports $|r^* - r_{\text{measured}}|$.

C. Layer 3: Statistical tail bounds on formal robustness

Safety requirements are Signal-Temporal-Logic (STL) formulas with quantitative robustness ρ (Fainekos–Pappas semantics; $\rho > 0$ = satisfied with margin) [13]. The challenge metric “% episodes in a safe equilibrium” is the special case $\Pr[\rho \geq 0]$; we report the full ρ distribution and its tails. The scored metric is estimated empirically by the classical arms; the Q-arm’s certified bound (Sec. V) is its conservative companion, stated with its direction.

IV. THE COMMUTING-PAULI INTERACTION COMPILER

Each cross-modal pairwise interaction term is a Pauli string $P_a = \sigma_{a_1} \otimes \dots \otimes \sigma_{a_k}$ with coefficient c_a (learned or inherited); the interaction tensor is $T = \sum_a c_a P_a$. Grouping into commuting families and per-family Clifford diagonalization are standard VQE measurement-reduction machinery [11], [12]; our contribution is their use as an exact, prunable compiler for cross-modal interaction tensors. The compile pass:

Exactness (E-comp-1): $\|T_{\text{exact}} - T_{\text{compiled}}\| \leq 10^{-6}$ relative (numerics/coefficient-estimation only). Runtime cost per family scales with the diagonal evaluation, not pairwise $O(N^2)$ enumeration — the attention bottleneck answered with an identity plus an $O(N \cdot k)$ -style diagonal kernel. Pruning is

compositional, with Eq. (3): the certified pruning budget. **Scope and complexity, stated honestly.** The Pauli table is not extracted from an arbitrary dense matrix (that would require up to 4^k terms); it is extracted from the *compressed interaction accumulators* of Sec. VII, whose bounded rank bounds the term count, and families beyond a top- k by $\|c\|_1$ are pruned with a certificate-bound on the tail (R5). The grouping is greedy graph coloring — polynomial, not optimal — and the realized family count m is a *measured* output of N7, reported with its pruning certificate. We claim *no classical computational speedup* from the Clifford step: on classical hardware the rewrite is an exact algebraic identity whose value is exactness, certifiable pruning, and the hardware pathway; on quantum hardware, simultaneous measurement in a common eigenbasis is the standard VQE shot-reduction device that motivates commuting grouping, and k-NoCliD’s non-commuting relaxations [12] mark the adjacent frontier. The quantum content is the object language — Pauli observables, Clifford frame, measurement basis — not a speed claim.

Why this survives the QASA objection. QASA showed a capacity-matched classical low-rank bottleneck matches a PQC’s error metrics [9]. Our claim is not “the encoding is powerful” but “the compiler is exact, certifiable, and executable”: generic feature crosses have no commuting-group structure, and Performer/FAVOR+ [10] approximates the softmax kernel with distributional error and no notion of a family — it cannot prune with a printed certificate, and it emits no hardware table.

V. SAFETY EVALUATION ENGINE

A. Oracle

The Sec. 5.2 seeded-scenario requirement makes the simulator a reversible deterministic function of the seed:

$$O_{\text{true}} : \text{seed} \mapsto \text{scenario} \mapsto \text{rollout}(\pi) \mapsto \rho \mapsto [\rho < 0], \quad (6)$$

the object the classical arms evaluate. Amplitude estimation’s end-to-end value turns entirely on the oracle it can realize, so we state that explicitly. A coherent oracle that evaluates a *neural* rollout inside the circuit would carry policy and dynamics as reversible subcircuits — prohibitive depth on hardware and exponential state-vector cost on simulators. A coherent oracle that merely *loads* N pre-computed outcomes faces the $\Omega(N)$ state-preparation (QRAM) bottleneck. Our design avoids both horns:

- 1) *Structured domain, free preparation.* The seeded scenario set is a product domain (perturbation ranges per STL predicate, per the Sec. 5.2 sampling distribution), so the uniform superposition over $\log_2 |S|$ seed qubits is

TABLE III
THREE-ARM SAFETY ESTIMATION (PRE-REGISTERED STRUCTURE;
COMPLEXITY AT EMPIRICAL $p = 10^{-5}$; THE Q-ARM’S ESTIMAND IS
 p_{cert}).

Arm	Method	Query cost
Q	IQAE, Bayesian posterior [14], [15]	$\sim 10^2\text{--}10^3$ oracle queries $\times$ depth
C-strong	RESTART splitting + GEV tail fit [18]	$10^3\text{--}10^4$ rollouts (measured)
C-naive	Monte-Carlo counting	$\sim 10^6 + (\text{cost shown, not-run})$

$H^{\otimes \log_2 |S|}$ — no amplitude-encoding of a lookup table. Reversible arithmetic for the predicates adds $O(\log |S|)$ ancillas, so the full AE circuit uses $O(\log |S|)$ qubits and simulated state-vector cost is polynomial in $|S|$ ($|S|^{O(1)}$) — tractable at our scenario counts under the challenge’s simulation mandate.

- 2) *Closed-form certificate-margin predicates.* The Q-arm oracle computes the *certified* verdict $\tilde{p}(s) = [\text{margin}_{\text{cert}}(s) < 0]$ (fusion joint 3): Layer-1 margins are arithmetic over core norms and box geometry; Layer-2a margins are the exported per-box SOS constants. Both are small reversible arithmetic/comparison circuits of printed size — never a neural rollout; the policy is deliberately outside the circuit.
- 3) *Conservative direction.* Sound certificates give $\Pr[\text{true failure}] \leq \Pr[\tilde{p}(s) < 0]$, so the Q-arm estimates an upper bound on the scored quantity — the correct ISO 26262 object, and the direction is stated on every reported number.

Oracle circuit size (qubits, depth, gates) and the query budget are printed per benchmark, so the claim is auditable: a *query-complexity advantage under an explicitly priced, realized oracle*. If the certificate stack degrades (R2/R3), the cheap-predicate premise degrades with it and the classical arms carry the section. Layer 3 is *offline evaluation-time*: the ≤ 100 ms real-time budget governs the runtime path of Sec. II (compiled diagonal kernels + monitor), which never executes an amplitude-estimation loop.

B. The three-arm race

Amplitude estimation estimates p to absolute error ε with $O(1/\varepsilon)$ *idealized-oracle* queries versus Monte-Carlo’s $O(1/\varepsilon^2)$ — quadratic query-complexity savings on the rarest scored quantity, standard in the QAE literature [14], [15], [17]. We claim no end-to-end speedup a priori: the race measures it. The Grover depth $1/\sqrt{p}$ is real simulator cost at our budgets — precisely why the race is pre-registered (R4) and the claim never assumes quantum hardware. The arms estimate complementary quantities: Q estimates the certified (conservative) violation probability p_{cert} on the cheap oracle above; C-strong estimates the empirical failure probability from true rollouts. R4 is defined on a common estimand: RESTART+GEV is also run on the *certified* predicate itself, so “Q loses” means IQAE’s credible interval on p_{cert} is wider than RESTART’s on the same quantity at matched total budget (printed oracle cost included); the empirical tail

from true rollouts is reported alongside as the companion number. The crossover point (oracle complexity at which the classical arm wins) is a reported output of N6. If Q loses, we print that; Layers 1–2 stand independently. Adjacent QAE applications include structural reliability [15], flow-based failure sampling [16], and MPPI control [17]; Tabarraei and Das-Tanaka use lookup-table oracles over finite ensembles — precisely the case the oracle dilemma flags — and our in-domain delta is the realized cheap oracle over a structured domain fused with exact structural certificates.

C. Closure

Scenario optimization (Campi–Garatti): with N scenarios and solution complexity d , $\Pr[\text{violation} > \varepsilon] \leq \beta$ whenever $N \geq N(\varepsilon, \beta, d)$ — printed per claim as an (N, ε, β) triple [19]. **Conformal calibration**: all intervals receive split-conformal coverage correction on a held-out slice (never the training slice; leakage rule R12). **Falsification loop**: failures found by RESTART/IQAE feed back as training-slice augmentations, new long-tail slice entries, and monitor training positives; the report’s figure shows tail mass shrinking across rounds.

VI. RUNTIME MONITOR

A two-layer guard on the action stream with theorem-budgeted guarantees.

Layer 1 — PDU hygiene (zero extra FLOPs). The compiled commuting-family table (Sec. IV, pass 3/6) already computes a syndrome per inference step. If the syndrome changes between consecutive steps, the packet is gray-listed: this catches bit-flip corruption of the Pauli register — sensor dropouts, camera blinding, CAN-bus corruption — the exact failure modes of the challenge’s synthetic perturbation suite.

Layer 2 — Classical-shadow statistics. With M random projection directions $v_1 \dots v_M$ (Huang–Kueng–Preskill shadows [20]), median-of-means concentration over the scalars $s_i(x) = \langle G_i, x \rangle$ gives a statistically correct false-alarm budget at memory $O(M \cdot s)$ and a few matvecs of latency — continuous OOD shift that a parity bit cannot see. A split-conformal gate sets the user-visible false-alarm rate (e.g. $\leq 1\%$). Gray-list fallback: the largest-radius action from the Layer-1 certified safe set — the monitor never aborts into the void.

VII. TRAINING ENGINE

QTT reshape [6]: $m = \prod m_i$, $n = \prod n_j$ (binary or small-factor modes, bit-ordering per layer type; N2’ resolves the ordering). **TT-cross** [5], [7]: cores $G_1 \dots G_d$ are built from $O(\sum m_i n_i r_i^2)$ maxvol queries of W ’s entries, never forming W — SVD-free and VRAM-safe for 7B-scale layers.

Expressiveness, measured not assumed. Long-range multimodal attention is known to be low-rank-friendly (softmax kernels compress well at moderate rank), but we do not assume it: N2 registers a per-layer rank-vs-accuracy curve at the target ratios with a pre-registered bound (R1: $> 5\%$ drop at $2\times$ demotes the compression claim), and the bond allocator then

TABLE IV
EXPERIMENT MATRIX (PRE-REGISTERED; GPU-H ARE QUOTA-HOURS).

ID	Experiment
N1	AD backbone fine-tune + INT8 reference
N2	{TT,QTT}×TT-cross sweep, 6 bond plans × 2 backbones
N2'	Bit-ordering sensitivity (3 orders, 1 layer, 1 seed)
N3	Layer-1 certificate table (Lipschitz products + tightness ratio κ)
N4	Layer-2a SOS boxes (3 points × 2 tracks × slices)
N5	Layer-2b Lyapunov curve + bond-spectrum predictor
N6	Layer-3 safety suite (IQAE vs RESTART+GEV vs MC; oracle cost + crossover)
N7	Compiler exactness + prune certificates + latency
N8	(optional) PQC RL toy head + noise sweep
N10	Syndrome-shadow monitor demo
N11	ALS/DMRG-native fine-tune, both backbones
N12	Safety-critical last-pass on falsified slice
N13	Cross-track mirror (N2/N3/N5 on second backbone)
N14	Clean-env end-to-end bench audit

TABLE V
MAIN RESULTS TABLE (STRUCTURE; † = PRE-REGISTERED TARGET).

Row	Pillar	GPU-H	qcert-QTT	INT8	SV
Accuracy @ 2× (drop vs dense)	A	8	$\leq 5\%^\dagger$	measured	meas
Accuracy @ 4× (drop vs dense)	A	24	target $\leq 5\%^\dagger$	measured	meas
Certified-safe-set % @ 2×	A	2	$> 90\%^\dagger$	— (not computable)	—
Certified-safe-set % @ 4×	A	2	target $> 50\%^\dagger$	—	—
Latency (ms, stated profile)	A	3	≤ 100	measured	meas
	A	6			
	C	8			
	B	6			
	tertiary monitor engine	5			
	engine	8			
	engine	3			
	A	12			
1 exactness	all	4	$\ T_{\text{ex}} - T_{\text{comp}}\ \leq 10^{-6}$ rel.		
2 prune-certificate corr.			predicted vs measured ≥ 0.95		
3 latency			≤ 100 ms at full table		
4 hardware self-consistency			compiled vs simulated within tolerance		
Arm	Output				
Q (IQAE)	credible interval on $p_{\text{cert}} = \Pr[\text{margin}_{\text{cert}} < 0]$ (c)				
C-strong (RESTART+GEV)	profile-likelihood CI on p				
C-naive (MC)	cost shown; Clopper-Pearson CI				
Scenario-opt	(N, ε, β) triple				
Conformal	coverage on held-out slice				

spends rank where the *certified long-tail margin* is cheapest in certificate terms.

ALS/DMRG-native fine-tuning: each core is optimized directly (regularized least squares with exact multilinear gradients; AMEn enrichment modulates bond dimension where the allocator’s safety signal flags it). Memory scales with core size, not layer size. Training passes: P1 baseline fine-tune; P2 compress → ALS fine-tune (the deliverable); P3 safety-critical last-pass on the RESTART/IQAE-falsified failure slice only (tail-shrinkage evidence, ~hundreds of hard examples).

Safety-budgeted bond allocator: per layer,

$$r^* = \arg \min_r \max_{\text{slices}} (\text{certified long-tail margin at } r), \quad (7)$$

using (1) layer sensitivity to the certified-margin computation, (2) bond-spectrum entropy, (3) monitor hygiene bits. The long-tail slice is a design selector, not a report line.

VIII. EXPERIMENTAL SETUP

Tracks and baselines (Sec. 5.4): AD — LLaVA-1.5-7B (Alpamayo-adjacent head) on nuScenes [21]/CARLA-style slices, vs. INT8 bitsandbytes; robotics — MiniVLA-class backbone (MIT code, HF checkpoints; Open X-Embodiment lineage [22]) on LIBERO-style slices, vs. INT8. Strong extra baselines: SVD-at-matched-ratio, tuned Performer at matched capacity, RESTART+GEV (not naive MC). **Hardware:** Kaggle 2×T4 (30 GPU-h/wk, 120 h ceiling), local RTX 5060 for small runs; all budgets are quota-hours, silicon-hours logged separately. **Discipline:** mean ± std over ≥ 3 seeds; every report table has a ledger row (research/experiments.md); every experiment has a pre-registered kill criterion.

Budget rollup: 87 GPU-h core, 33 h headroom vs. the 120 h ceiling; weeks 1–6 schedule in 08-experiments.md. Week 1: N1+N2'+N3 (the cheapest exposure of the two load-bearing facts: does the baseline behave, and does QTT compress with certificates attached?). The optional N8 PQC demo uses a quantum natural policy gradient head [23] for illustration only; it is cut first under quota pressure (R10).

IX. RESULTS FRAMEWORK

Per challenge Sec. 5.5, all values are mean ± std over ≥ 3 runs. Tables below state the pre-registered targets; measured values land from the ledger during weeks 1–6 and replace the placeholders. No number in this report is fabricated; every number in the final report will be regenerable via `qicert.bench.<table>`.

X. ABLATION AND QUANTUM JUSTIFICATION

Per challenge Sec. 5.5, the QI-isolating ablation is mandatory and is designed to be the strongest section of the report.

A1 (compression×certificate): at matched ratio, INT8 matches accuracy but cannot report the certificate column — the ablation shows the asymmetry directly (the number that cannot be computed is the win). The tightness ratio κ is reported per point: the certificate column is conservative-but-quantified, and Layer 2a closes the gap locally.

A2 (compiler vs encoding): per-family leave-one-out with *predicted* error $\|\sum_{a \in F_j} c_a\|_1$ vs measured Δ ; and compiler-vs- encoding: the exact identity vs a bare feature cross (shows the family structure is what buys exactness, not the encoding label). Tuned Performer at matched capacity is the classical ceiling; winning it is upside, losing it does not kill the claim (the primary axis is exactness + certificates + hardware table).

A3 (IQAE vs strong classical): the three-arm race at matched query budget; if Q loses, the report prints that (R4) and the claim rests on Layers 1–2 and the strong classical arm.

A4 (monitor): syndrome-shadow vs autoencoder baseline: correction rate at matched overhead; false-alarm rate vs the shadow theorem bound ($\leq 3\times$ or demoted, R7).

A5 (training engine): ALS-in-compressed vs dense-then-compress at matched ratio on the safety slice ($\leq 2\%$ worse or demoted, R6).

XI. LIMITATIONS AND RISKS

Pre-registered kill/demote criteria (full register in 09-risks-kills.md): R1 compression accuracy $>5\%$ drop at $2\times \rightarrow$ CompactifAI-style softer claim + certificate story; R2 Layer-1 bound too loose ($<50\%$ certified safe set) $\rightarrow$ SOS becomes primary; R3 SOS SDP intractable $\rightarrow$ Lyapunov curve primary, SOS reported as “attempted, too slow”; R4 IQAE loses to RESTART $\rightarrow$ RESTART stands, IQAE = future work; R5 family count too large $\rightarrow$ top- k by $\|c\|_1$ with certificate-bound; R6 ALS-in-compressed $>2\%$ worse $\rightarrow$ demote to post-processing; R7 shadow FPR $>3\times$ theorem $\rightarrow$ PDU hygiene only; R8 allocator tradeoff $\rightarrow$ reported honestly; R9 cross-track non-comparable $\rightarrow$ pipeline claim only; R10 quota overrun $\rightarrow$ N8 cut first, then N13 reduced; R11 clean-env failure $\rightarrow$ report held; R12 conformal leakage $\rightarrow$ claim degraded until re-run.

Explicit framing disclaimers. We claim no classical speedup from the Clifford step (Sec. IV) and place amplitude estimation strictly offline with printed oracle costs (Sec. V); both are stated in-body so no stronger claim can be imputed. No end-to-end advantage over classical estimators is claimed for IQAE a priori — the pre-registered race at matched total budget and its measured crossover decide (Sec. V). Runtime parity with classical baselines is an acceptance target, not an assumption: E-comp-3 (≤ 100 ms at the full compiled table) and the N14 clean-env bench are the measured evidence.

Three things that cannot kill the core claim: (1) the exact Lipschitz-product certificate — a theorem, not a benchmark; (2) compilation-as-identity + pruning certificate — a mechanism, not a number; (3) the three-layer safety methodology — a formal spec plus raced estimators plus conformal closure, not a score. The final table is built from whatever layers survive their criteria; the package always runs and reports something certified.

Certification framing: the STL specs map to ISO 26262 / IEC 62061 / ISO 10218 artifacts (hazard $\rightarrow$ safety goal $\rightarrow$ STL formula $\rightarrow$ robustness predicate $\rightarrow$ certified tail probability), and the certificate stack provides the evidence trail an assessor expects.

XII. RESOURCE DECLARATION

Hardware: Kaggle $2\times$ NVIDIA T4 16 GB (primary), local RTX 5060 (secondary). **Total:** 87 GPU-h core (quota-hours) + headroom; silicon hours logged separately in the ledger. **Simulation:** CUDA-Q (pinned version), PennyLane, torch — versions in environment.yml. **Shot/gate budgets:** per benchmark, stated in the compiler’s pass-5 table. **Energy:** measured by the qicert profiler (kWh + CO₂e at declared grid intensity). Reproducibility evidence: qicert.bench.all from a fresh virtualenv, verified by N14.

XIII. CONCLUSION

We submit objects the baselines structurally cannot produce: exact certificates that are functions of the compressed cores, an exact compilation identity with per-family provable pruning, quadratic query-complexity savings on the rarest scored quantity under a realized, priced oracle, and a package that reproduces every table from a clean environment. All claims are either theorems, structural properties, or pre-registered measurements with kill criteria — the honesty contract of Sec. 5.5 operationalized. The full argument, theory, and risk register are maintained in the submission folder of the qicert repository.

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